



Trinity College Dublin

Coláiste na Tríonóide, Baile Átha Cliath

The University of Dublin

Power Consumption Analysis of Containerised 5G Deployments with Variable Topologies and Load

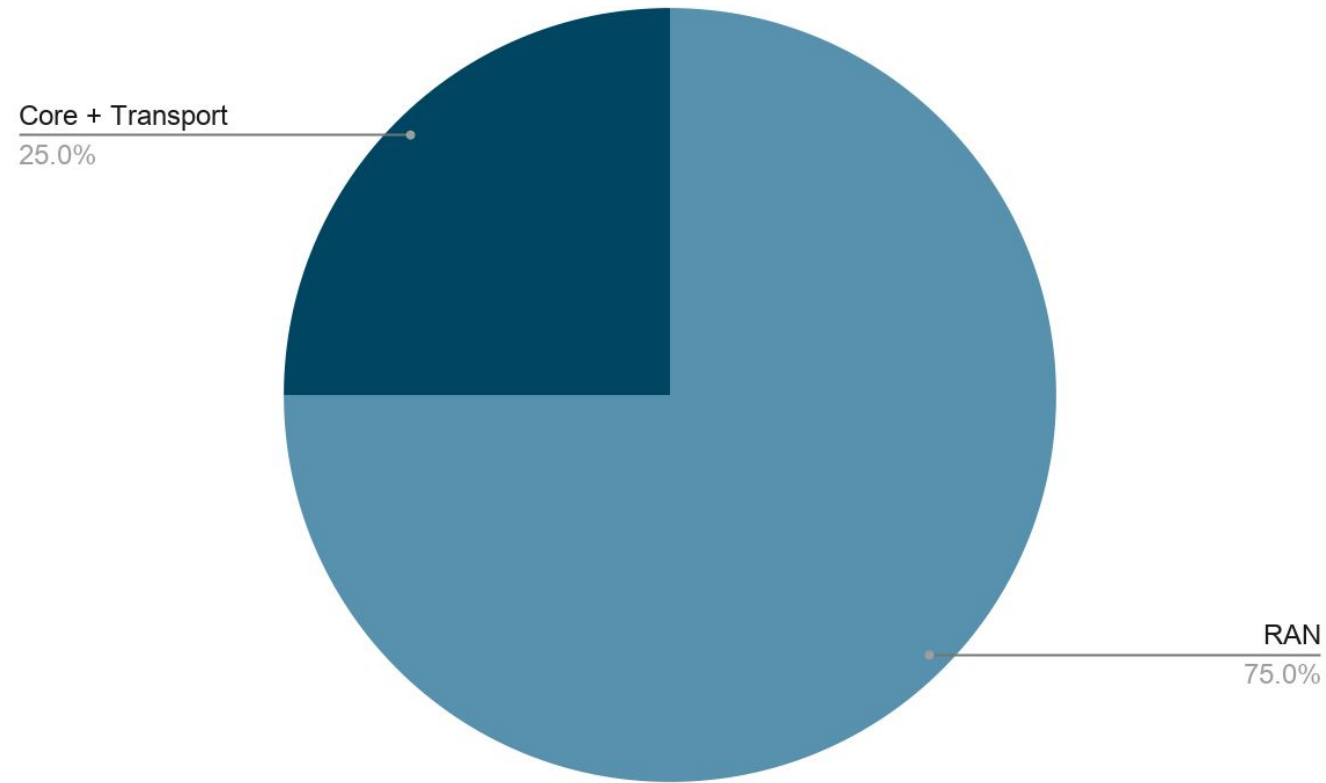
MSc Computer Science - Future Networked Systems

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Motivation

- Radio Access Network (RAN) consumes over 75% of mobile network energy
- Densification keeps adding base stations, raising the overall footprint
- At scale, small per-site gains compound into large savings



Source: Gudepu et al., 2024

Research Questions

Main

RQ1: How does DU count affect power consumption, and is a centralised CU more energy-efficient than multiple CUs serving the same DUs?

Supporting

RQ2: How does traffic load (UE count and CQI) influence power consumption within each topology?

RQ3: Can a threshold-based dynamic scaling policy reduce energy waste by matching the number of active DUs to current traffic demand?

Background

- 5G RAN is disaggregated into CU, DU and RU (3GPP TS 38.401), which allows many deployment configurations
- The same DUs can be served by one centralised CU or several distributed ones, and each arrangement has a different energy cost
- These costs depend on topology, load and placement, and aggregate power figures cannot separate them: per-process measurement is needed

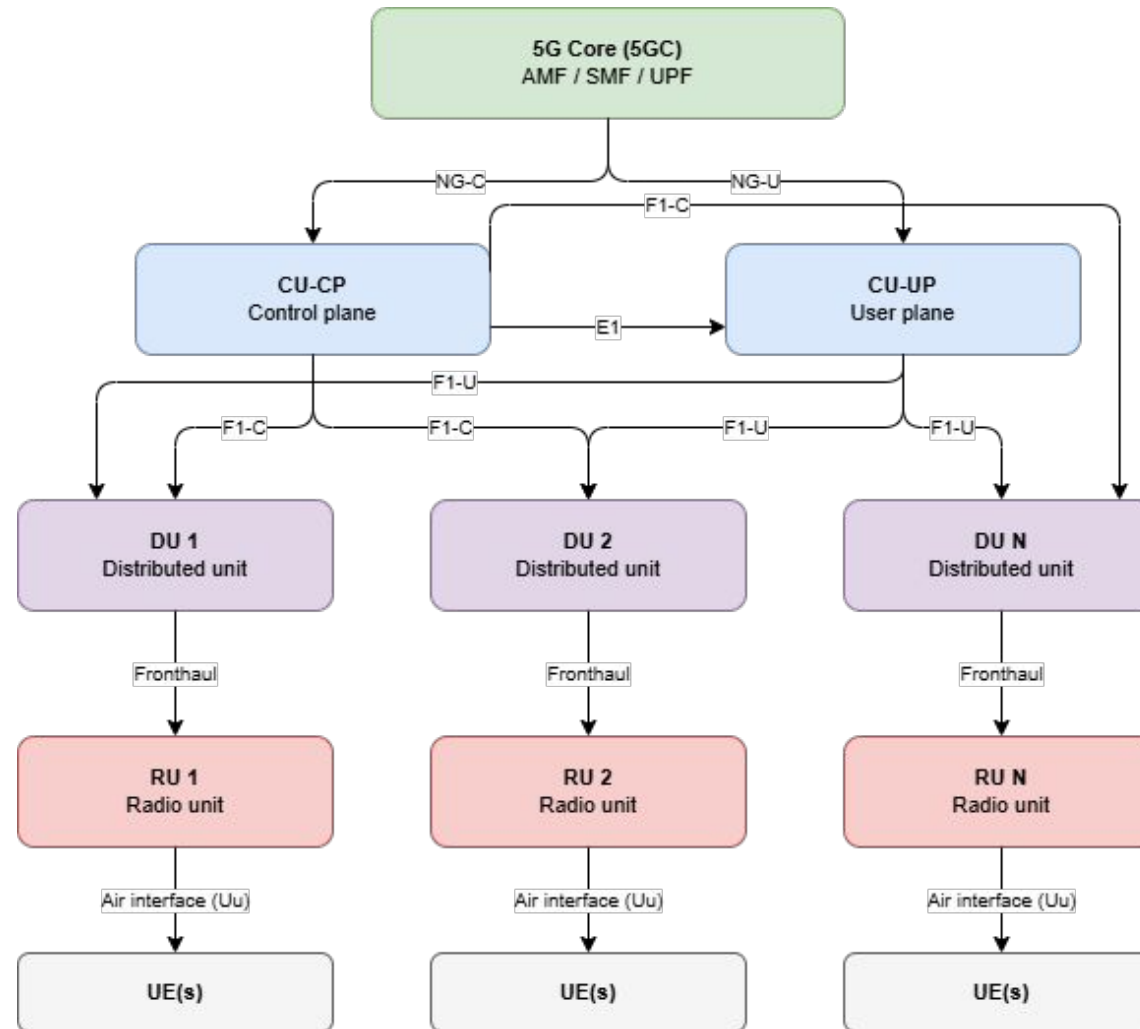
Related Work

Paper	Contribution	Gap addressed
EARNEST (Gudepu et al., 2024)	Per-component power across Monolithic, Disaggregated, CUPS RAN	Fixed topology only
Al-Tahmeesschi et al. (2025)	Real SDR hardware, Split 7.2b and 8	No topology variation
Centofanti et al. (2024)	Scaphandre vs Kepler vs s-tui comparison	No RAN context
Bellin et al. (2023)	Container vs VM vs bare-metal 5G core	Core only, no RAN
This work	Variable CU/DU topology, ZMQ real traffic, 1185 validated runs across two campaigns	Variable topology + real traffic + process-level breakdown

System & Experimental Design

Open5GS core, srsRAN CU/DU, and per-process power via Scaphandre

System Overview



Full end-to-end containerised 5G stack: Open5GS core, srsRAN CU-CP/CU-UP/DU, srsUE via ZMQ, power monitored per process via Scaphandre/RAPL

Experimental Setup

Testbed:

- CloudLab bare-metal node: AMD EPYC 7452, 32 cores, 125 GB RAM
- srsRAN Project: CU-CP, CU-UP, DU (one container per component)
- Open5GS 5G Core (constant across all experiments)
- srsRAN 4G (srsUE) via ZMQ, 10 MHz bandwidth

Power Measurement:

- Scaphandre + RAPL: per-process power via Linux cgroup
- Prometheus + InfluxDB + Telegraf
- Sampling interval: 5 seconds
- 5 runs per configuration

Measurement Methodology

Power measurement:

- Scaphandre reads RAPL energy counters every 5 seconds
- Per-process attribution via Linux cgroup: one cgroup per container
- Three phases per run: idle, DL active (iperf3), UL active (iperf3)

Data collection:

- Per-run CSV: timestamp, phase, component, PID, watts
- Automated validation: unique UE IPs, non-zero DL/UL, all components present
- Throughput: iperf3 logs per UE, aggregated across all UEs

Experiment Matrix

ru_dummy (Docker testmode):

- 1CU × {1..8} DU, {1,4,16,64,96} UE, CQI {5,10,15}
- 1125 runs, no real UE traffic
- Focus: topology and load scaling at scale

ZMQ (real UE traffic):

- Multi-DU: 1CU × {1..4} DU, {1,4} UE/DU, 5 runs - 40 runs
- Multi-CU: 2CU × {1,2} DU/group, {1,4} UE/DU, 5 runs - 20 runs
- Real throughput + per-component power

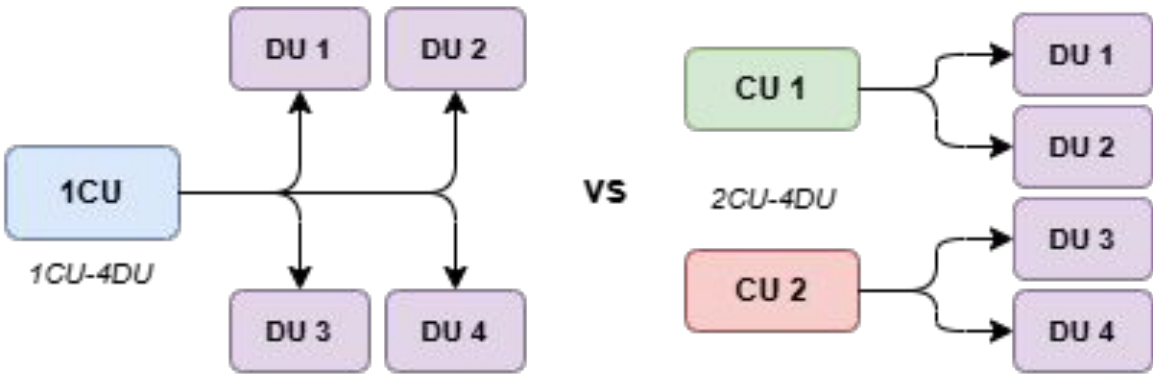
Experiment Topologies

ru_dummy (Docker testmode)

Vertical scaling (1CU)

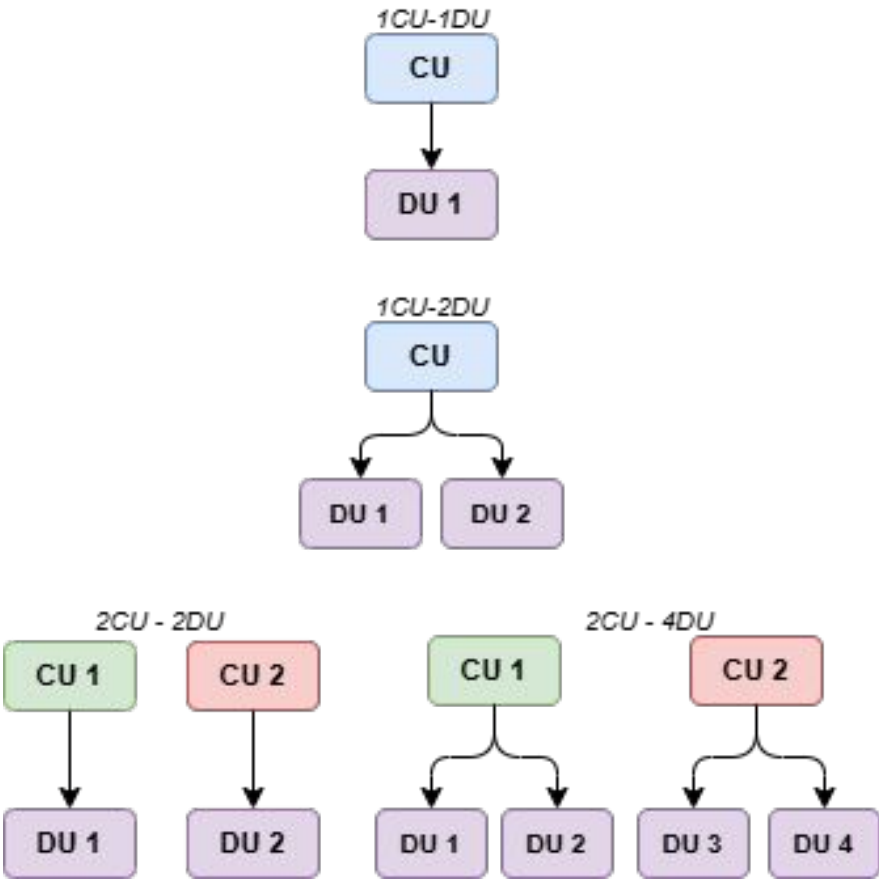


Horizontal scaling (multi-CU), example at 4 DU total



Also tested: 3CU-6DU, 4CU-4DU, 2CU-6DU, 2CU-8DU, 3CU-3DU

ZMQ (real UE traffic)



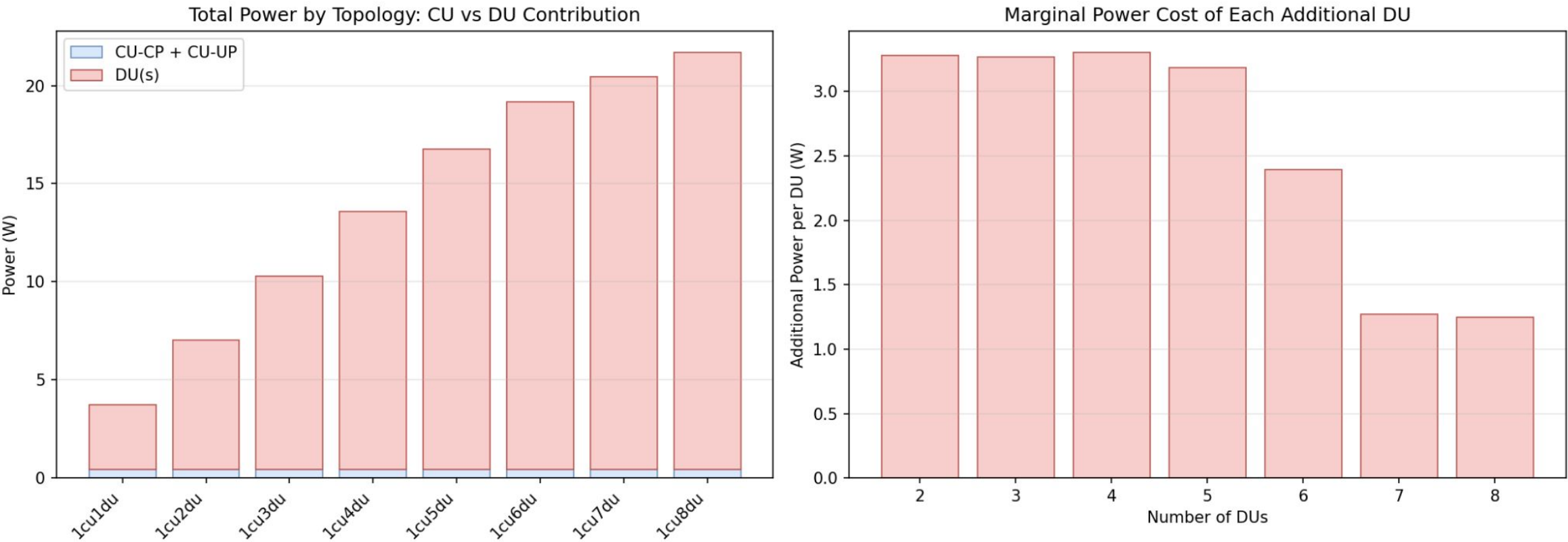
Results

Power breakdown, topology scaling, and centralisation across both campaigns

ru_dummy: Key Results

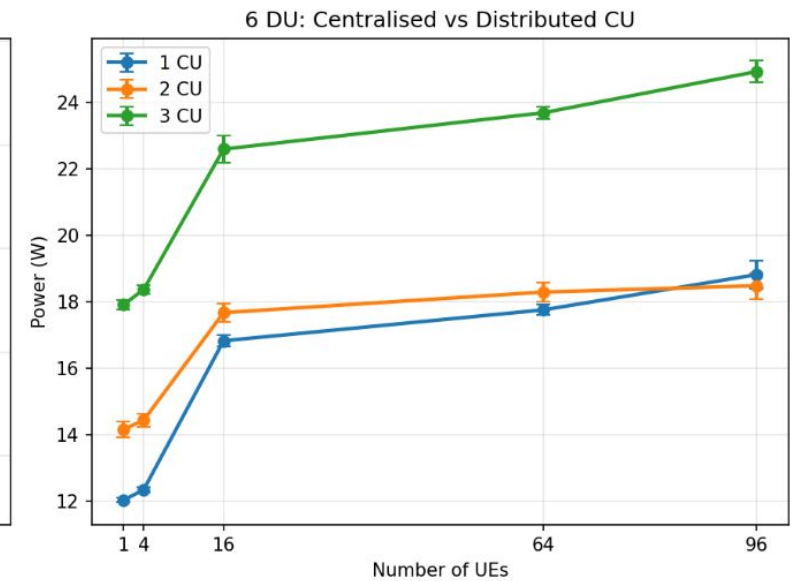
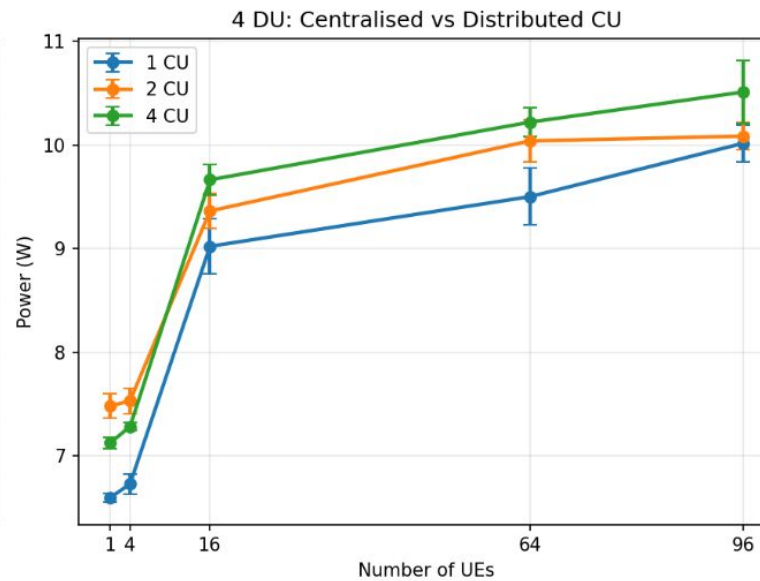
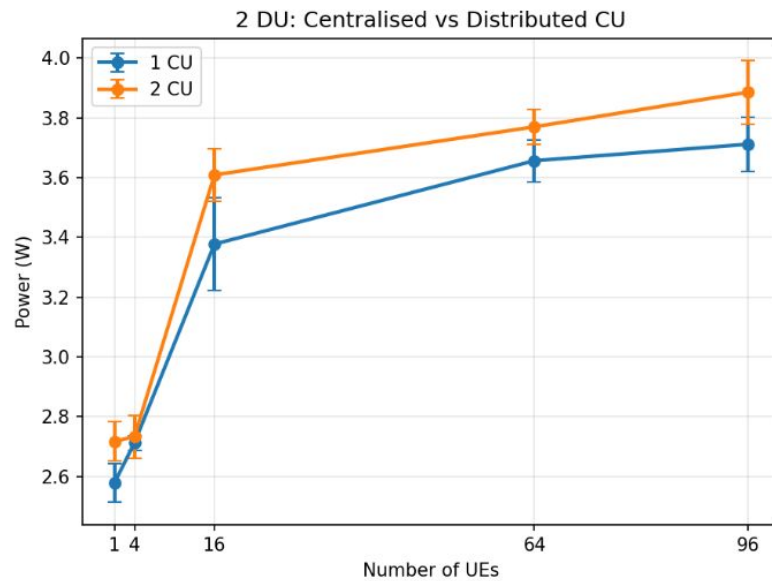
- CU overhead fixed at 0.42W regardless of DU count
- DU scaling linear up to 5 DUs (~3.2W per additional DU); saturation beyond DU6
- CQI has negligible impact across all topologies and UE counts
- Largest power jump between 4 and 16 UEs
- At equal DU count, centralised 1CU consistently draws less power than distributed configurations

ru_dummy: Power Breakdown by Topology



ru_dummy: Centralised vs Distributed CU

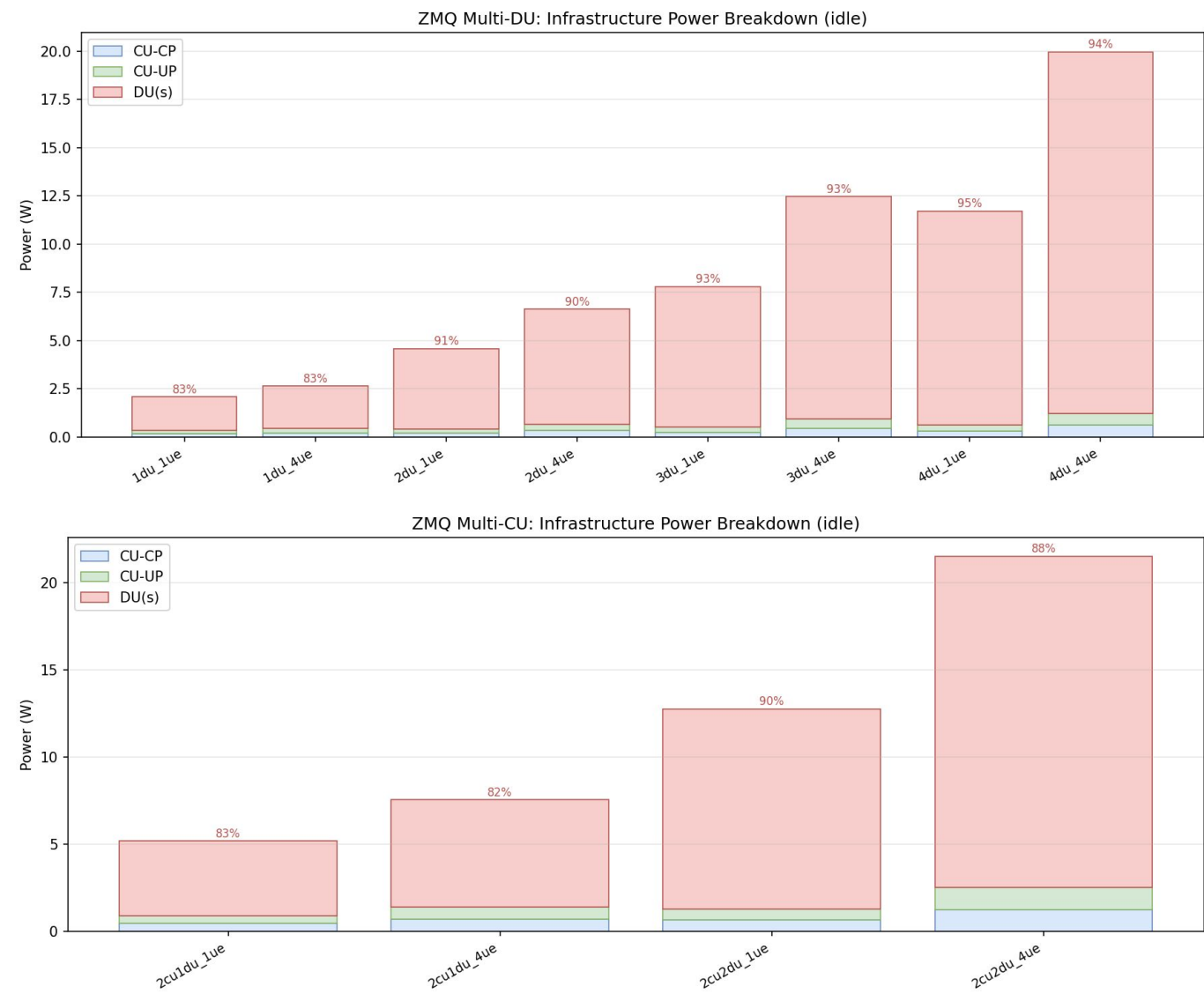
Power Consumption: Centralised vs Distributed CU (CQI=10)



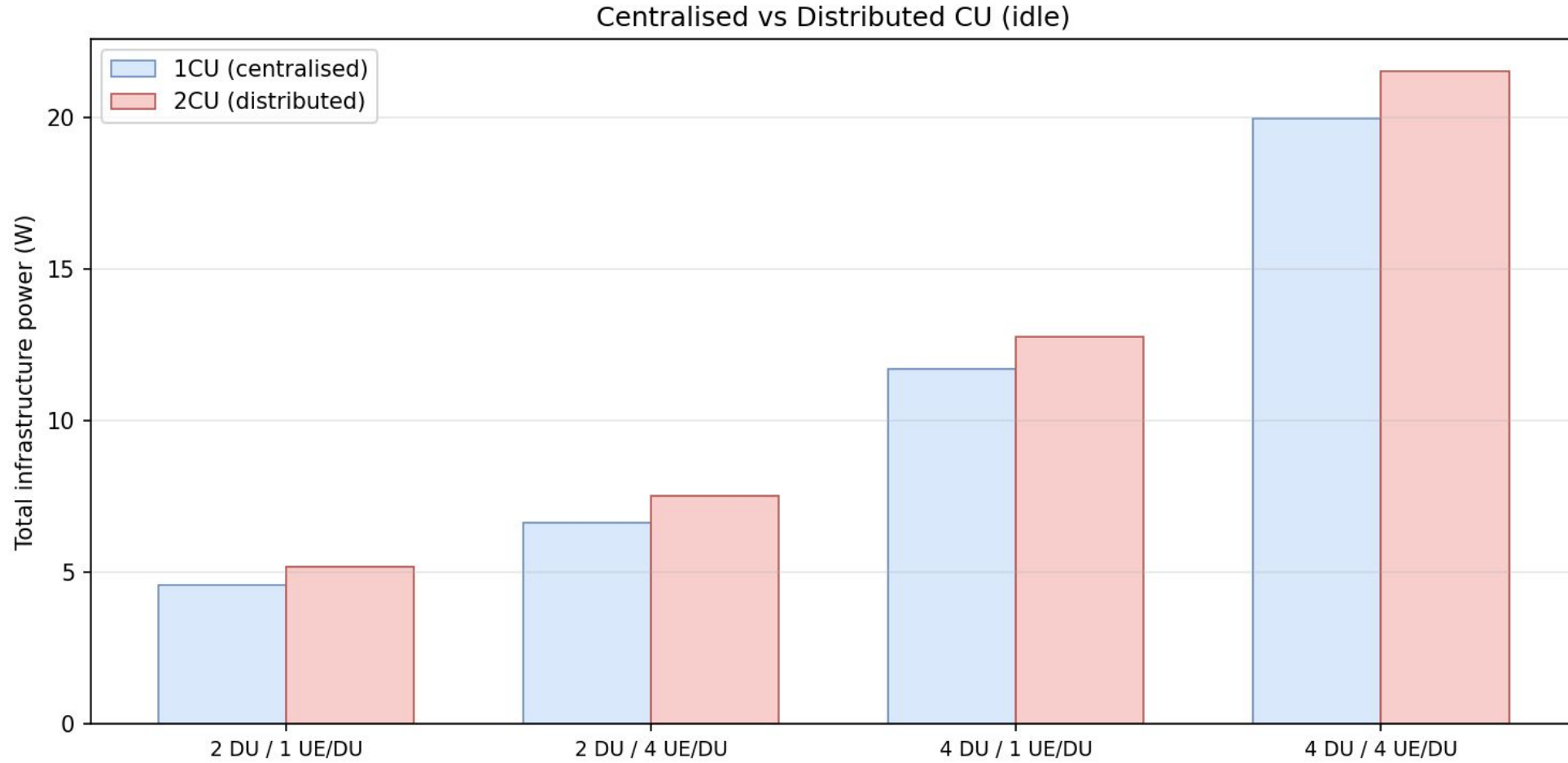
ZMQ: Key Results

- DU dominates infrastructure power (83-94%), consistent with ru_dummy results
- Fixed power accounts for 70-80% of per-unit DU consumption
- CU-CP and CU-UP scale linearly with DU count
- 2CU overhead: 5-12% vs equivalent 1CU, decreasing with scale
- Energy efficiency: 0.10 W/Mbps at 1CU-1DU, up to 0.30 W/Mbps at 1CU-4DU with 4 UE/DU

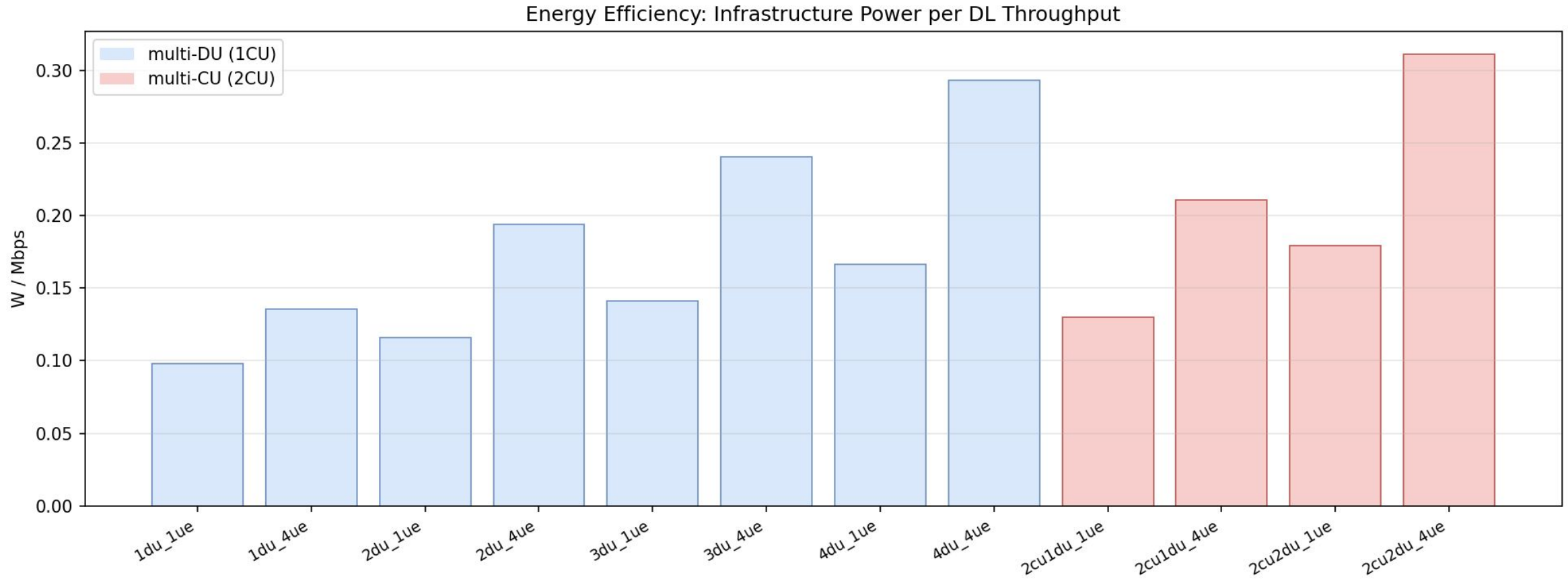
ZMQ: Infrastructure Power Breakdown



ZMQ: Centralised vs Distributed CU

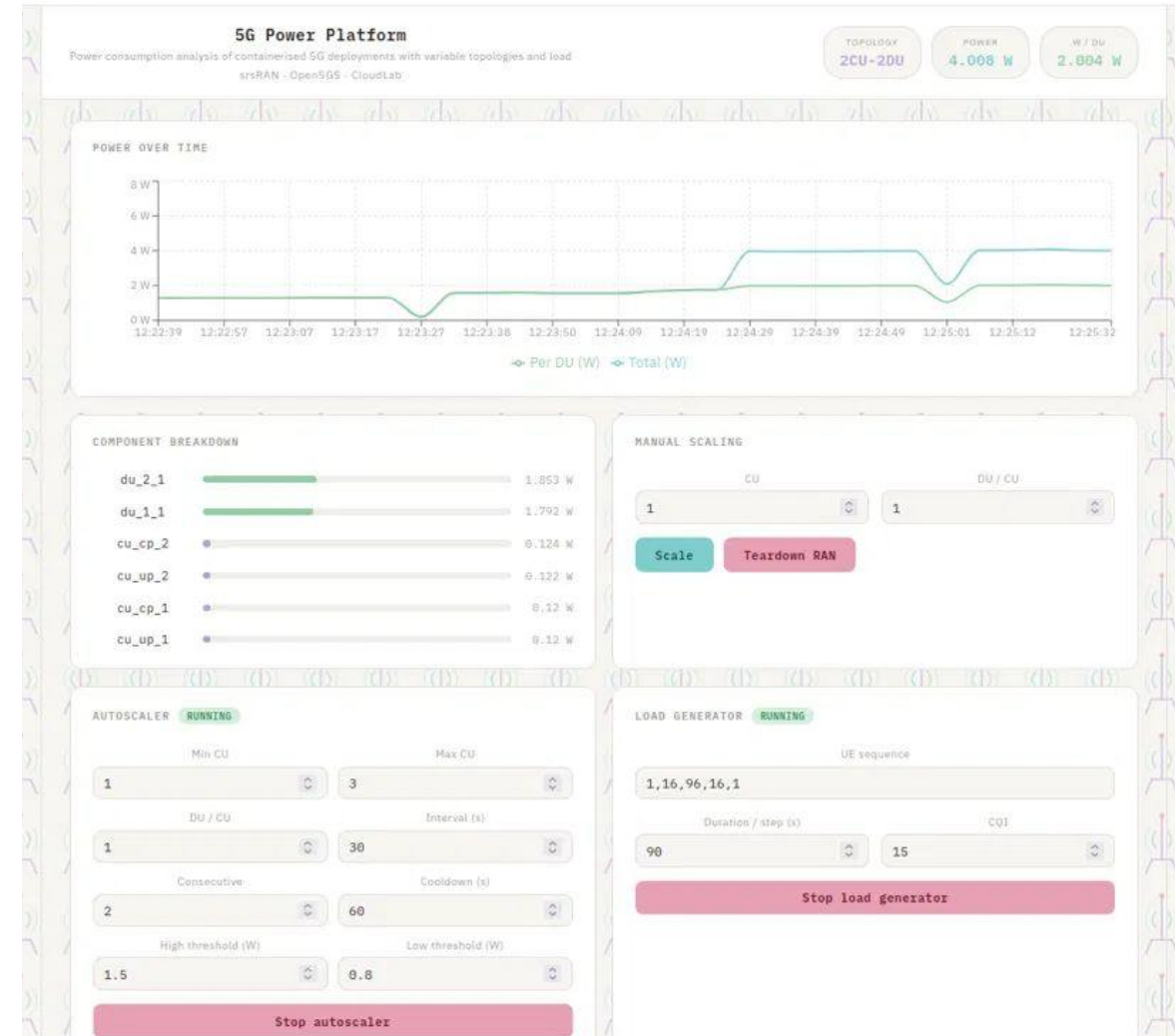


ZMQ: Energy Efficiency



Orchestration Platform & Dynamic Scaling

- Threshold-based autoscaler on per-DU power: configurable thresholds, cooldown, consecutive triggers
- Runtime config generation from templates: topologies are not predefined
- Per-process attribution via Scaphandre/RAPL feeds the control loop
- Addresses RQ3: dynamic topology reconfiguration to reduce energy waste



Key Findings

The centralisation trade-off

- Distributing the CU costs a small, fixed per-CU overhead (5-12% in ZMQ), with no crossover in the measured range
- The overhead amortises across the DUs each CU serves, not across the total: fewer DUs per CU means a higher penalty
- Energy is one of four decision axes, and the only one quantified here: latency, capacity and resilience push towards distribution

Wrapping Up

What the data shows, its limits, and where it points next

Limitations

- Single bare-metal node: centralised and distributed CU share the same physical host
- RAPL measures CPU and DRAM only: NIC consumption from F1/E1/NG traffic not captured
- ZMQ broker single-process: scheduling asymmetry at higher UE loads, max 4 UE/DU
- ru_dummy does not generate real user-plane traffic
- Dynamic scaling validated on ru_dummy track only

These limits act uniformly across topologies: comparative findings hold, absolute values are conservative

Future Work

- ML-based autoscaler for proactive topology reconfiguration based on traffic patterns
- Validate autoscaler under real traffic by extending dynamic scaling to ZMQ track
- Multi-node deployment and Kubernetes cluster for true geographical CU-DU separation
- Extend ZMQ matrix to higher DU counts via distributed broker architecture
- Validate findings on hardware RF testbed using SDR frontend

Conclusions

- **RQ1:** DU count is the primary determinant of RAN power; at equal hardware, energy favours centralisation (see trade-off)
- **RQ2:** Load matters far less than topology.
Fixed power is 70-80% of per-DU consumption; CQI impact is negligible
- **RQ3:** Threshold-based autoscaler demonstrates runtime topology reconfiguration, validated on ru_dummy
- Process-level attribution via Scaphandre/RAPL is viable in containerised deployments, addressing a gap in the literature
- The biggest lever is switching off idle DUs, not optimising active ones



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Thank you

Questions?

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github.com/martafra/5g-power-dissertation